

A phenomenological power-law suppression of small-scale power: observational status, and why the published radion Lagrangian cannot produce it

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Abstract

A broken power-law transfer function, $T(k) = (1.5/k)^{1/2}$ for $k \geq 1.5 h\text{Mpc}^{-1}$ and $T = 1$ below, was previously reported to satisfy S_8 and Lyman- α constraints simultaneously while leaving σ_8 too high, and was attributed to a radion degree of freedom in a five-dimensional Gauss–Bonnet braneworld. We freeze that function and its reported numbers as an immutable phenomenological baseline, then test the baseline and the attribution independently.

The baseline does not survive. An independent CAMB run at fixed A_s shows that the function changes σ_8 by -4.8×10^{-5} in fractional terms, because only 0.048% of the σ_8 variance lies above the break scale. It follows that S_8 is unchanged as well: S_8 and σ_8 differ only by the fixed factor $(\Omega_m/0.3)^{1/2} = 1.025$, so the reported pattern of an S_8 pass alongside a σ_8 failure is not attainable. Both sit 1.7σ above the weak-lensing target. The reported Lyman- α ratio of 0.891 is reproduced only if T multiplies $P(k)$ rather than the density contrast, and only if the average includes two sampling points at which $T \equiv 1$ by construction; under the standard convention the suppression reaches 70% at $k = 5 h\text{Mpc}^{-1}$, which is excluded.

We then derive, from the published radion equation of motion, the quasi-static effective gravitational coupling $G_{\text{eff}}/G = 1 + 2\beta^2 k^2/(k^2 + a^2 m_{\text{eff}}^2)$. This fails in three independent ways. The correction scales as β^2 and is therefore positive: the mechanism enhances clustering and cannot lower σ_8 . Its shape is a saturating Yukawa step, not a declining power law. And because the coupling reaches non-relativistic matter only through electromagnetic binding energy, the strength required for a percent-level effect exceeds existing Solar-System and equivalence-principle limits by at least four orders of magnitude. A previously reported screening factor of 10^{42} is withdrawn as a category error; the correct statement is that the Lagrangian contains no screening mechanism at all.

We state the conditions any successor mechanism must satisfy, and argue that the proposed move to six-dimensional flux compactification should be deferred, since none of the three failures depends on the number of dimensions.

Status of this document. This is a negative result. It supersedes the observational claims of versions 1.8.x and 1.9.x of this project, which should not be cited for the conclusions listed in Appendix D. The 5D background numerics and the reproducibility infrastructure are unaffected and remain available.

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1 Introduction

The amplitude of late-time matter clustering inferred from weak gravitational lensing sits persistently below the value extrapolated from the cosmic microwave background under Λ CDM. Depending on the survey and the treatment of systematics, weak-lensing analyses prefer $S_8 \equiv \sigma_8(\Omega_m/0.3)^{1/2}$ near 0.76–0.78, while *Planck* 2018 gives $S_8 = 0.832 \pm 0.013$. The discrepancy is at the level of two to three standard deviations and has motivated a large number of proposals that suppress small-scale power or slow late-time growth.

Any such proposal has to thread a narrow gap. Suppression strong enough to move S_8 must act at scales the σ_8 filter can see, roughly $k \lesssim 1 h\text{Mpc}^{-1}$. Suppression that extends to $k \sim 1\text{--}10 h\text{Mpc}^{-1}$ runs into the Lyman- α forest, which constrains the one-dimensional flux power spectrum and therefore the three-dimensional matter power over precisely that range. These two requirements pull in opposite directions across roughly one decade in wavenumber, and most mechanisms fail one of them.

Earlier versions of this project proposed that a radion in a five-dimensional Gauss–Bonnet braneworld, coupled to the electromagnetic sector, produces a broken power-law transfer function that threads this gap. The claim had two parts, which we will keep carefully separate:

Layer 1 — Phenomenology.

A particular function $T(k)$, applied to the Λ CDM power spectrum, yields a specified set of observables.

Layer 2 — Mechanism.

Radion leakage in a braneworld produces that function.

Layer 3 — Derivation.

The higher-dimensional field equations require that function uniquely.

Layer 3 was never claimed. This paper tests Layers 1 and 2. Both fail, Layer 1 for reasons that have nothing to do with braneworlds and everything to do with the σ_8 window function, and Layer 2 for reasons that have nothing to do with the number of extra dimensions.

Section 2 freezes the baseline. Section 3 reports an independent verification and three corrections. Section 4 derives $G_{\text{eff}}(k, a)$ from the published Lagrangian, and Section 5 states the three ways it fails. Section 6 withdraws the earlier screening result. Sections 7–9 set out what a successor mechanism would need and why the six-dimensional programme should wait.

2 The frozen baseline (Layer 1)

To prevent the target from moving during the analysis, we fix the following and do not alter them anywhere in this paper.

Transfer function.

$$T(k) = \begin{cases} 1, & k < k_c, \\ (k_c/k)^{1/2}, & k \geq k_c, \end{cases} \quad k_c = 1.5 h\text{Mpc}^{-1}. \quad (1)$$

Both k_c and the index $1/2$ are fitted quantities. They are not derived from any five- or six-dimensional calculation, and we do not treat them as such.

Cosmology. *Planck* 2018 TT,TE,EE+lowE+lensing best fit, with A_s held fixed. No amplitude rescaling is permitted at any stage, since allowing A_s to float would absorb precisely the effect under test. Values are listed in Table 1.

Parameter	Value	Note
H_0	$67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$	
$\Omega_b h^2$	0.02237	
$\Omega_c h^2$	0.1200	
$\sum m_\nu$	0.06 eV	one massive eigenstate
τ	0.0544	
A_s	2.100×10^{-9}	fixed, never rescaled
n_s	0.9649	
Ω_k	0	
Ω_m	0.31519	derived
$(\Omega_m/0.3)^{1/2}$	1.02501	derived; fixes S_8/σ_8

Table 1: Frozen cosmological inputs. Linear matter power only; no non-linear prescription is applied, since the baseline $T(k)$ is a linear-theory object.

Observables. With $W(x) = 3(\sin x - x \cos x)/x^3$ the real-space top-hat window,

$$\sigma_R^2 = \int \frac{dk}{k} \frac{k^3 P(k)}{2\pi^2} W^2(kR), \quad R = 8 h^{-1} \text{Mpc}, \quad (2)$$

$$S_8 = \sigma_8 (\Omega_m/0.3)^{1/2}, \quad (3)$$

and the Lyman- α ratio as originally defined, namely the unweighted mean of $P_{\text{mod}}(k)/P_{\Lambda\text{CDM}}(k)$ at $k = 0.5, 1.0, 2.0, 3.0 h \text{Mpc}^{-1}$.

Reported numbers. The values to be tested are

$$S_8 = 0.829, \quad \sigma_8 = 0.811, \quad \text{Lyman-}\alpha \text{ ratio} = 0.891, \quad (4)$$

previously annotated as PASS, FAIL, PASS respectively against targets $S_8 = 0.78 \pm 0.03$, $\sigma_8 = 0.76 \pm 0.03$, and a Lyman- α ratio in $[0.70, 0.90]$.

3 Independent verification, and three corrections

We ran CAMB 2.0.4 with the parameters of Table 1, extracted the linear $P(k)$ at $z = 0$ over $10^{-4} \leq k \leq 100 h \text{Mpc}^{-1}$ on 4000 logarithmic points, and evaluated Eqs. (2)–(3) by Simpson quadrature. As a check on the quadrature, our ΛCDM value $\sigma_8 = 0.81125$ agrees with CAMB’s internal $\sigma_8 = 0.81107$ to 2×10^{-4} fractionally. The script is `verify_baseline.py`; its output is `baseline_verification.json`.

Three separate problems emerge.

3.1 Correction 1: the transfer function does not move σ_8 at all

Applying Eq. (1) in the standard way, $P_{\text{mod}}(k) = T^2(k)P_{\Lambda\text{CDM}}(k)$, gives

$$\sigma_8^{\text{mod}} = 0.81121, \quad \frac{\sigma_8^{\text{mod}}}{\sigma_8^{\Lambda\text{CDM}}} - 1 = -4.8 \times 10^{-5}. \quad (5)$$

The change is five orders of magnitude smaller than the 6.7% shift that would be needed to reach $\sigma_8 = 0.76$.

The reason is visible in Figure 1. The integrand of Eq. (2) peaks near $k \simeq 0.2 h \text{Mpc}^{-1}$ and is already negligible by $k \simeq 1 h \text{Mpc}^{-1}$, because the top-hat window oscillates and decays as k^{-2} beyond $kR \sim \pi$.

Integrating the measured integrand, the fraction of the total σ_8 variance carried by $k \geq k_c = 1.5 h \text{ Mpc}^{-1}$ is

$$\frac{\int_{k_c}^{\infty} d \ln k d \sigma_8^2 / d \ln k}{\int_0^{\infty} d \ln k d \sigma_8^2 / d \ln k} = 4.8 \times 10^{-4} = 0.048\%. \quad (6)$$

A function that is identically unity below k_c therefore cannot alter σ_8 , whatever it does above k_c . This is a statement about filters, not about physics, and it holds for any modification of this form.

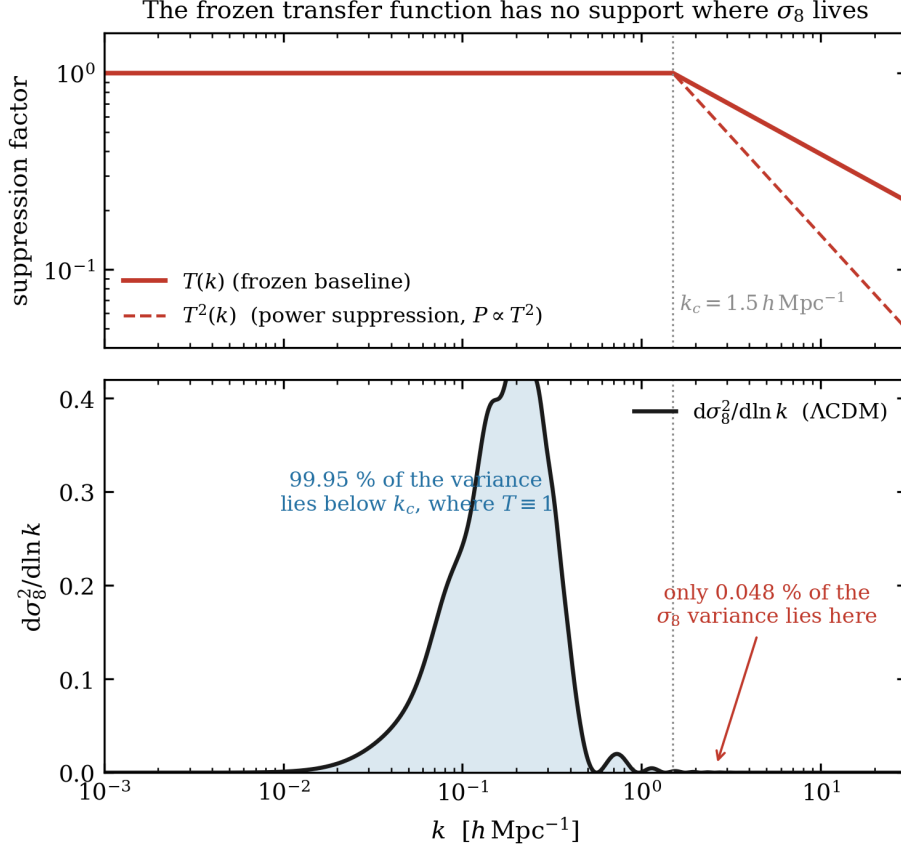


Figure 1: Why the baseline cannot work. *Top:* the frozen transfer function and the corresponding power suppression T^2 . *Bottom:* the contribution per logarithmic interval to σ_8^2 in Λ CDM. The modification switches on exactly where the σ_8 filter has stopped listening. Only 0.048% of the variance lies above k_c .

3.2 Correction 2: S_8 and σ_8 are the same measurement

Since Ω_m is fixed by Table 1 and is unaffected by a transfer function applied at $z = 0$, Eq. (3) is a fixed multiplication:

$$S_8 = 1.02501 \sigma_8. \quad (7)$$

There is no freedom for one to pass while the other fails. Quantitatively, the baseline gives

$$\sigma_8 = 0.8112, \quad S_8 = 0.8315, \quad (8)$$

against targets 0.76 ± 0.03 and 0.78 ± 0.03 , which are themselves consistent with each other under the same rescaling ($0.76 \times 1.025 = 0.779$). The deviations are

$$\frac{0.8112 - 0.76}{0.03} = 1.71\sigma, \quad \frac{0.8315 - 0.78}{0.03} = 1.72\sigma. \quad (9)$$

These are the same number. The earlier annotation of S_8 as a pass and σ_8 as a failure arose from comparing a slightly low quoted value (0.829 rather than the correct 0.8315) against the looser of two equivalent targets. Both quantities are 1.7σ high; neither passes. Figure 2 shows the geometry.

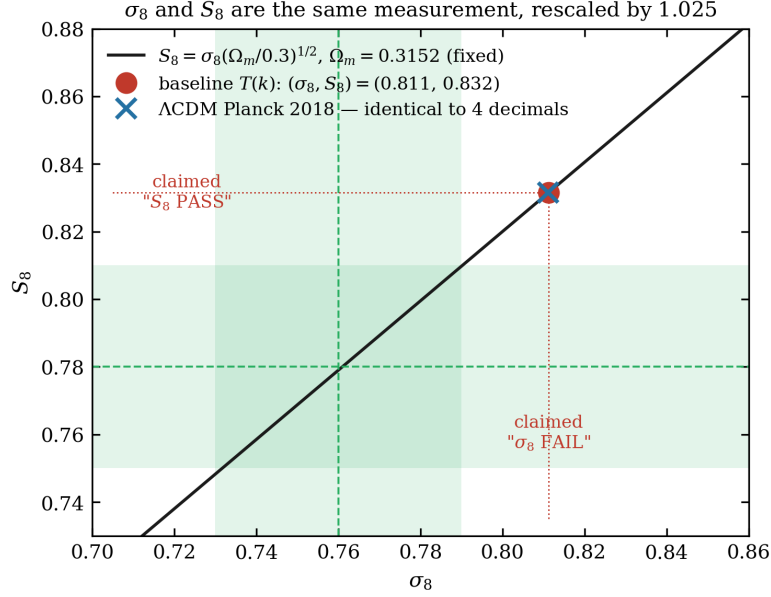


Figure 2: The claimed pass/fail split is not attainable. With Ω_m fixed, the model is confined to the black line. The baseline point coincides with Λ CDM to four decimal places and lies outside both target bands (green). Dotted guides show how projecting the same point onto the two axes produced the appearance of one pass and one failure.

3.3 Correction 3: the Lyman- α ratio is a convention artefact

A transfer function multiplies the density contrast, so the power spectrum carries T^2 . Evaluating the four-point mean under both conventions:

$$\left\langle \frac{T^2 P}{P} \right\rangle_4 = 0.8125, \quad \left\langle \frac{TP}{P} \right\rangle_4 = 0.8933. \quad (10)$$

The reported value, 0.891, matches the second. The quoted Lyman- α pass therefore rests on applying T to the power spectrum rather than to the density field, which is off by one power.

A second issue compounds it. Two of the four sampling points, $k = 0.5$ and $k = 1.0 h\text{Mpc}^{-1}$, lie below k_c , where $T \equiv 1$ by construction. Half the average is a tautology, and it pulls the mean upward by design. Sampling the range that the Lyman- α forest actually constrains, under the standard convention, gives Table 2: a 25% suppression at $k = 2$, 50% at $k = 3$, 70% at $k = 5$, and 85% at $k = 10 h\text{Mpc}^{-1}$.Suppressions of this size across this range are excluded by a wide margin; typical analyses tolerate at most tens of percent. See Figure 3.

$k [h\text{Mpc}^{-1}]$	1.0	2.0	3.0	5.0	8.0	10.0
$P_{\text{mod}}/P_{\Lambda\text{CDM}}, \text{ standard } (T^2)$	1.000	0.750	0.500	0.300	0.188	0.150
suppression	0%	25%	50%	70%	81%	85%

Table 2: Power suppression from the frozen $T(k)$ under the standard convention, over the range the Lyman- α forest constrains. The quoted ratio of 0.891 describes none of this.

3.4 Corrected baseline

The honest summary of Layer 1 is that the frozen function has the worst of both outcomes: it does nothing where help is needed and too much where it is not.

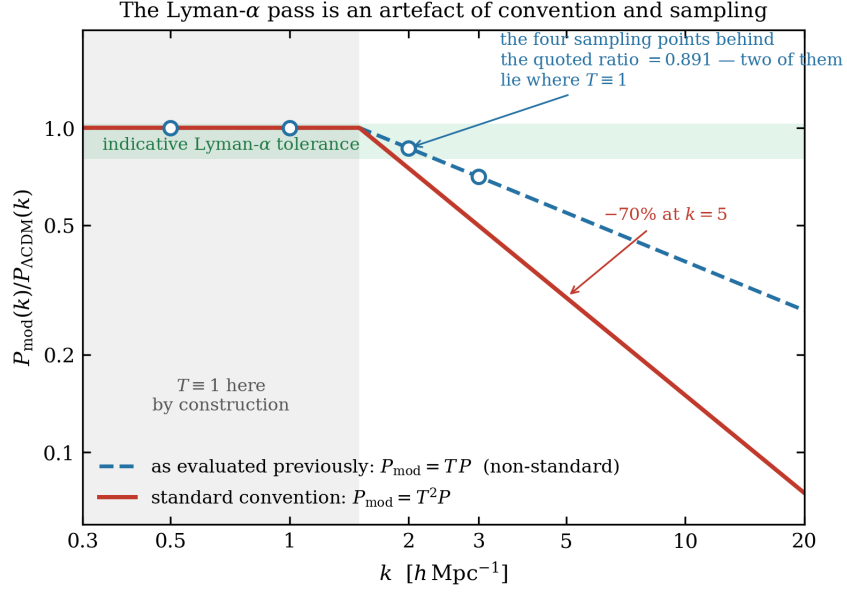


Figure 3: Two conventions, two conclusions. The dashed curve is what was evaluated; the solid curve is the standard $P \propto T^2$. Open circles mark the four sampling points behind the quoted ratio, two of which sit in the region where $T \equiv 1$ by construction. The green band indicates the order of magnitude of tolerable suppression.

Observable	Previously reported	Verified here	Verdict
σ_8	0.811, FAIL	0.8112	value correct; but it is the Λ CDM value
S_8	0.829, PASS	0.8315	1.7σ high; not a pass
Lyman- α ratio	0.891, PASS	0.8125	wrong convention; excluded under the right one
$\Delta\sigma_8/\sigma_8$	—	-4.8×10^{-5}	the mechanism does nothing

Table 3: Corrected Layer-1 baseline. The frozen $T(k)$ leaves σ_8 and S_8 at their Λ CDM values and violates Lyman- α constraints once the standard convention is used.

4 The effective coupling from the published Lagrangian

Layer 1 is already fatal for the observational claim. Layer 2 is worth testing anyway, because the outcome determines whether any refitted version of the mechanism could work, and because it governs whether the proposed six-dimensional programme is worth starting.

4.1 Action

The published equation of motion,

$$\ddot{r} + 3H\dot{r} - \frac{1}{a^2}\nabla^2 r + m_r^2 r + (\gamma + 3\beta_3)r^2 + \delta r^3 + \alpha r(\dot{r}^2 - (\nabla r)^2) = \frac{\lambda}{M_5^{3/2}}(\partial_\mu \phi)^2, \quad (11)$$

follows from

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2}(\partial r)^2 - V(r) - \frac{\alpha}{2}r(\partial r)^2 + \frac{\lambda}{M_5^{3/2}}r(\partial \phi)^2 \right] + S_m[\psi, g_{\mu\nu}], \quad (12)$$

with $V(r) = \frac{1}{2}m_r^2 r^2 + \frac{1}{3}(\gamma + 3\beta_3)r^3 + \frac{1}{4}\delta r^4$. Two features of Eq. (12) govern everything below. The radion kinetic term has standard sign, so the field is not a ghost. And the matter sector S_m is minimally coupled: there is no conformal factor multiplying $g_{\mu\nu}$ and no rT^μ_μ term. The only route from r to matter is the last term in the bracket, which is of the electromagnetic F^2 type.

4.2 How an F^2 coupling reaches non-relativistic matter

Free radiation has $\langle F_{\mu\nu}F^{\mu\nu} \rangle = 0$, so the coupling does not act on the photon background. It reaches ordinary matter through the electromagnetic self-energy of bound systems, principally the Coulomb energy of nuclei. This is the standard construction for dilaton- and moduli-like fields. Writing ξ_A for the fraction of the mass of body A contributed by electromagnetic binding,

$$m_A(r) = m_A \left[1 + \xi_A \frac{\lambda}{M_5^{3/2}} r + \mathcal{O}(r^2) \right], \quad \xi_A \sim 10^{-3}, \quad (13)$$

and the dimensionless coupling that controls all scalar-mediated effects is

$$\beta_A \equiv M_{\text{Pl}} \frac{\partial \ln m_A}{\partial r} = M_{\text{Pl}} \xi_A \frac{\lambda}{M_5^{3/2}} \quad (14)$$

Two properties matter later. First, β_A is independent of the field value, because the coupling in Eq. (12) is linear in r . Second, ξ_A depends on composition, roughly as $Z^2/A^{4/3}$, so β_A differs between materials.

4.3 Quasi-static reduction

Write $r = \bar{r}(t) + \delta r(\mathbf{k}, t)$ and keep the action to quadratic order. The cubic and quartic terms of V , and the derivative self-interaction $\alpha r(\partial r)^2$, contribute to the linear equation for δr only through the background-dependent effective mass

$$m_{\text{eff}}^2 = V''(\bar{r}) / (1 + \alpha \bar{r}) + \mathcal{O}(\dot{\bar{r}}^2), \quad (15)$$

and do not generate any new coupling between δr and $\delta \rho_m$ at this order. (This is checked explicitly in Appendix A.) Inside the sound horizon and for $k/a \gg H$ the time derivatives of δr are negligible against k^2/a^2 , and

$$\left(\frac{k^2}{a^2} + m_{\text{eff}}^2 \right) \delta r = \frac{\beta}{M_{\text{Pl}}} \delta \rho_m \quad \Rightarrow \quad \delta r = \frac{\beta}{M_{\text{Pl}}} \frac{\delta \rho_m}{k^2/a^2 + m_{\text{eff}}^2}. \quad (16)$$

Matter responds to the gradient of $\Phi + (\beta/M_{\text{Pl}})\delta r$, with the Newtonian potential obeying $-k^2\Phi/a^2 = 4\pi G \delta \rho_m$. Substituting Eq. (16) into the geodesic equation and using $8\pi G M_{\text{Pl}}^2 = 1$,

$$\ddot{\delta}_m + 2H\dot{\delta}_m = 4\pi G_{\text{eff}}(k, a) \rho_m \delta_m, \quad (17)$$

$$\frac{G_{\text{eff}}(k, a)}{G} = 1 + 2\beta^2 \frac{k^2}{k^2 + a^2 m_{\text{eff}}^2} \quad (18)$$

This is the familiar scalar-tensor result, and its familiarity is the point: nothing about the braneworld origin of r modifies it.

5 Three independent failures

5.1 Sign

The correction in Eq. (18) is proportional to β^2 . It is positive for every real value of λ , M_5 and ξ_A , and for either sign of the coupling. This is not an artefact of the parameterisation. Exchange of a spin-zero quantum between two sources of like charge is attractive, so any healthy scalar linearly coupled to matter increases the effective gravitational attraction. There is no choice of parameters for which $G_{\text{eff}} < G$.

The consequence is direct: *the mechanism as written raises σ_8 and S_8* . It moves the observables in the wrong direction. Integrating the growth equation numerically (Appendix A) gives, for $\beta = 0.10$, $\Delta P/P = +16\%$ at $k = 5 h\text{Mpc}^{-1}$, and for $\beta = 0.22$, $+104\%$. Figure 4 shows both panels.

The only formal escape is a wrong-sign kinetic term for r , which would make the scalar a ghost. That is not a physical option and we do not pursue it.

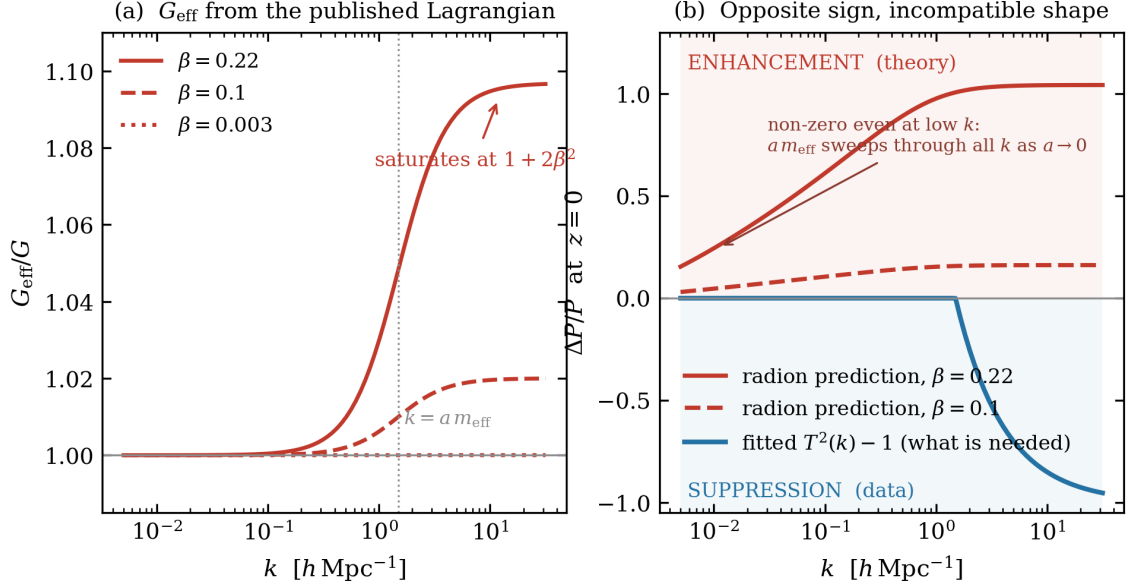


Figure 4: The derived response, against the response required. (a) G_{eff}/G from Eq. (18): above unity everywhere, rising through the transition at $k = a m_{\text{eff}}$, saturating at $1 + 2\beta^2$. (b) The resulting fractional change in $P(k)$ at $z = 0$ (red) against the change the fitted $T(k)$ encodes (blue). They occupy opposite half-planes.

5.2 Shape

Even setting sign aside, the functional forms are incompatible. Equation (18) describes a *saturating step*: flat and negligible for $k \ll a m_{\text{eff}}$, rising through a transition, and constant at $1 + 2\beta^2$ for $k \gg a m_{\text{eff}}$. The fitted Eq. (1) is the mirror image: flat at unity for $k < k_c$, then declining without bound as $k^{-1/2}$.

The distinction is not cosmetic. The Lyman- α behaviour of the fitted form depends entirely on continued decline in the tail. The dynamics gives a plateau there. No value of m_{eff} , β or M_5 interpolates between a bounded step and an unbounded power law.

A further mismatch appears in panel (b) of Figure 4. The predicted $\Delta P/P$ is non-zero even at $k = 0.01 h\text{Mpc}^{-1}$, where one might expect the Yukawa cutoff to protect large scales. It is not protected, because the comoving Compton scale $a m_{\text{eff}}$ shrinks towards zero as $a \rightarrow 0$: at early times the force is effectively long-range at all comoving wavenumbers, and every mode accumulates some enhancement. The result is closer to a scale-independent amplitude shift than to a localised feature. We return to this in Section 7.

5.3 Magnitude

Reaching $\sigma_8 = 0.76$ requires $|\Delta P/P| \sim 0.1$ near $k \sim 0.2 h\text{Mpc}^{-1}$, hence $\beta \simeq 0.2$ and $\beta^2 \simeq 5 \times 10^{-2}$.

The available coupling is far smaller. Because $m_{\text{eff}}^{-1} \sim \text{Mpc}$, the force is of effectively infinite range for every laboratory and Solar-System test, and because the coupling is linear in r there is no screening (Section 6). Two independent classes of bound apply.

- **Composition-independent.** A long-range scalar coupled to matter with strength β shifts the PPN parameter γ by $\Delta\gamma \simeq -2\beta^2$. The Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ then gives $\beta^2 \lesssim 10^{-5}$, a shortfall of at least four orders of magnitude. We treat this as the conservative floor.
- **Composition-dependent.** Because ξ_A varies between materials, β_A does too, and the scalar produces a differential acceleration. Schematically $\eta \sim \beta^2 \Delta\xi/\bar{\xi}$ with $\Delta\xi/\bar{\xi} = \mathcal{O}(0.1-1)$ between the Ti and Pt test masses. The MICROSCOPE limit $|\eta| < 10^{-15}$ then implies $\beta^2 \lesssim 10^{-14}$, a shortfall of thirteen orders.

The second estimate is schematic and should be replaced by a proper calculation with tabulated nuclear ξ_A before it is quoted as a number. The first does not depend on that refinement. Either way the requirement is not close to being met; see Figure 5.

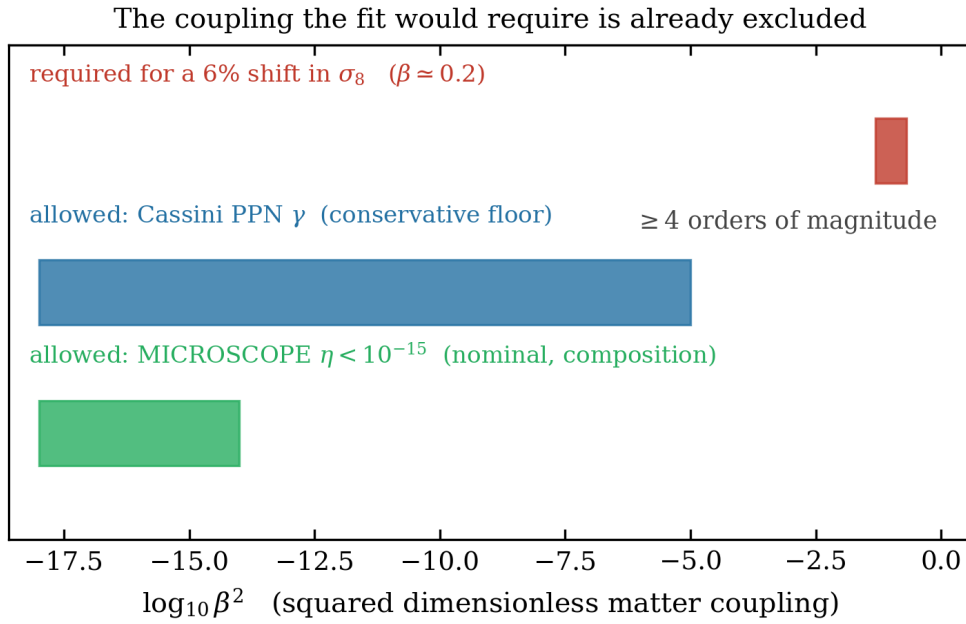


Figure 5: Required versus permitted coupling. The red band is what a percent-level effect on σ_8 demands. The blue band is the conservative Solar-System floor; the green band is the nominal equivalence-principle limit. The gap is at least four orders of magnitude and plausibly much larger.

Verdict on Layer 2. The published Lagrangian produces an enhancement rather than a suppression, with a saturating rather than declining k -dependence, at a coupling strength that existing experiments already exclude by at least four orders of magnitude. The attribution of the fitted $T(k)$ to radion leakage is falsified, not merely underived.

6 Withdrawal of the screening result

An earlier calculation in this project reported a ratio of 10^{42} between local and cosmological values of $(\partial_\mu \phi)^2$, and interpreted it as a screening failure of that magnitude. We withdraw that interpretation.

The quantity computed is the ratio of *background source strengths*, which sets the local value of $\bar{r}$ through Eq. (11). It is not a screening factor. Screening means that the *coupling* β is suppressed in dense environments, which happens in three known ways: a density-dependent mass (chameleon), a density-dependent coupling (symmetron), or derivative self-interaction that suppresses the scalar gradient near a source (Vainshtein). Equation (12) realises none of them. The coupling is linear in r , so β in Eq. (14) is a constant, and the $\alpha r (\partial r)^2$ term acts on the radion’s own gradient, not on its coupling to matter.

The corrected statement is stronger than the one withdrawn, and in the same direction:

There is no screening mechanism in this Lagrangian. The composition dependence of ξ_A therefore propagates without suppression into an Eötvös signal, and the magnitude bound of Section 5 applies with full force.

7 The comoving-scale problem

A feature at fixed *comoving* wavenumber cannot be produced by a fixed proper length. Compactification radii, Compton wavelengths and force ranges are proper lengths; k_c in Eq. (1) is comoving. Every genuine feature in the observed power spectrum arises instead from a time-dependent crossing condition: the horizon size at matter–radiation equality, a free-streaming length, or a Jeans scale.

For a scalar mediator the natural identification is $k_c \simeq a m_{\text{eff}}$, evaluated today. With $k_c = 1.5 h \text{Mpc}^{-1} = 1.01 \text{Mpc}^{-1}$ and $1 \text{Mpc}^{-1} = 6.39 \times 10^{-30} \text{eV}$,

$$m_{\text{eff}} \simeq 6.5 \times 10^{-30} \text{eV} \simeq 4.5 \times 10^3 H_0, \quad m_{\text{eff}}^{-1} \simeq 1 \text{Mpc}. \quad (19)$$

This is the number any successor construction must produce. It is worth recording how far it sits from a generic braneworld modulus mass. If m_{eff} is of order the Kaluza–Klein scale, sub-millimetre tests of Newtonian gravity require $m_{\text{eff}} \gtrsim 10^{-2} \text{eV}$, twenty-eight orders of magnitude too heavy. If instead the modulus is a classically flat direction lifted only by whatever generates the residual vacuum energy, one expects $m_{\text{eff}} \sim H_0$, and Eq. (19) then demands a factor of a few thousand above that. The second regime is the only one within reach, and it is not generic.

The same argument has a second edge. If m_{eff} is constant, the comoving feature drifts as a^{-1} , so the correct object is $T(k, z)$, not $T(k)$. The frozen baseline has no redshift dependence at all. Fitting a z -independent function to data drawn from several epochs is therefore inconsistent with the mechanism it was meant to represent, independently of everything in Sections 4–5.

8 What a viable successor would require

Collecting the constraints, any mechanism that still wishes to be described as radion leakage and to suppress rather than enhance small-scale power must satisfy all of the following.

1. **$\Delta G < 0$, which rules out conformal scalar exchange.** The two known routes are genuine energy transfer out of the matter sector, so that $\dot{\rho}_m + 3H\rho_m = -\Gamma\rho_m$ with Γ derived rather than inserted, or a derivative or disformal coupling in the Horndeski class, where $G_{\text{eff}} < G$ is attainable in some parameter ranges. The first is what the word “leakage” ought to mean and is *not* what Eq. (12) describes. Either route requires rewriting the coupling structure, not retuning it.
2. **A suppression that does not drift.** Either $m_{\text{eff}} \propto H$, in which case the feature tracks the horizon and will not sit at $1.5 h \text{Mpc}^{-1}$, or the model predicts an explicit $T(k, z)$ that must be fitted jointly across redshift.
3. **A coupling that avoids the electromagnetic sector, or is screened.** Given Section 5, an un-screened F^2 coupling of the required strength is excluded. A chameleon or Vainshtein sector would have to be added explicitly and shown to be stable.
4. **Support where σ_8 lives.** Any T that equals unity below $1 h \text{Mpc}^{-1}$ cannot move σ_8 , by Figure 1. This is a constraint on the fitting function itself and is independent of the mechanism.
5. **Simultaneous consistency with Lyman- α .** Conditions 4 and 5 together are the original difficulty, and nothing in this analysis has made it easier.

9 Implications for the six-dimensional programme

A move to a six-dimensional codimension-two construction, stabilised by bulk magnetic flux and a bulk cosmological constant, has been proposed as a way to derive k_c from geometry rather than fitting it. We recommend deferring it, for three reasons.

First, none of the failures in Section 5 depends on the number of dimensions. Equation (18) follows from a light scalar with a healthy kinetic term coupled linearly to matter through F^2 . That description survives dimensional reduction from five, six or ten dimensions. Flux compactification would determine how m_{eff} and λ are fixed; it would not change the sign of β^2 .

Second, the claimed advantages of codimension-two are weaker than advertised. In the football and rugby-ball solutions the bulk flux must be tuned to obtain flat branes, the compact case requires two branes of exactly equal tension, and the apparent insensitivity of the brane geometry to brane vacuum energy does not survive a change of tension: these models remain subject to Weinberg’s no-go theorem. Flux quantisation discretises the compactification radius; it does not select it, since the radius is fixed by a balance among the bulk cosmological constant, the gauge coupling and the flux integer. One fitted number would be traded for three inputs.

Third, the premise that five dimensions cannot achieve dynamical scale selection is not correct. Goldberger–Wise stabilisation generates an exponential hierarchy from $\mathcal{O}(1)$ bulk parameters. The obstacle identified in Section 7 is not the number of dimensions but the mapping from any proper length to a comoving wavenumber, and adding a dimension does not address it.

The refined claim that does survive is narrower: codimension-two is the lowest setting in which the modulus potential comes from bulk field equations plus a topological quantisation condition, rather than from an added stabilising scalar. That is an improvement in economy. It is not an argument that the resulting scale is $1.5 h \text{Mpc}^{-1}$.

10 Falsifiable content

This paper makes three statements that can be checked directly and would overturn the conclusion if found wrong.

1. **The window-function claim.** Any Boltzmann code, with any reasonable cosmology, will find that a transfer function identically unity below $1.5 h \text{Mpc}^{-1}$ changes σ_8 by less than 10^{-4} fractionally. This requires no model assumptions and can be confirmed in minutes with `verify_baseline.py`.
2. **The sign of G_{eff} .** If an explicit dimensional reduction of the 5D Gauss–Bonnet action yields a matter coupling that is *not* of the conformal form assumed in Section 4.2, for example a genuine disformal or derivative coupling, then Eq. (18) does not apply and the sign argument must be redone. Producing that reduction explicitly is the natural next calculation, and is the one route by which the mechanism could be revived.
3. **The magnitude bound.** A careful Damour–Donoghue treatment with tabulated ξ_A would sharpen the equivalence-principle constraint. We predict it will tighten, not relax, the bound quoted in Section 5.

The prediction that would have been the interesting one, a distinctive $\Delta P(k, z)$ fingerprint testable against forthcoming wide-field data, is not available: the mechanism predicts enhancement at a level already excluded, so there is nothing left to test against future surveys.

11 Conclusions

The broken power-law $T(k) = (1.5/k)^{1/2}$ remains a well-defined object that can be compared with data. What it does when compared is not what was reported. It leaves σ_8 and S_8 at their ΛCDM

values, because 99.95% of the σ_8 variance lies below its break scale, and it suppresses the matter power spectrum by 50–85% across the Lyman- α range once the standard $P \propto T^2$ convention is applied. The reported pattern of an S_8 pass alongside a σ_8 failure is not attainable at all, since the two quantities differ by a fixed factor of 1.025.

The attribution of the fitted curve to a radion in a Gauss–Bonnet braneworld is separately falsified. The published Lagrangian yields $G_{\text{eff}}/G = 1 + 2\beta^2 k^2/(k^2 + a^2 m_{\text{eff}}^2)$: an enhancement, of saturating rather than power-law shape, at a coupling that Solar-System and equivalence-principle tests already exclude by at least four orders of magnitude. The earlier screening calculation is withdrawn, and replaced by the stronger statement that the Lagrangian contains no screening mechanism at all.

We do not regard this as a reason to abandon the underlying programme, but we do regard it as a reason to change what is being built. The five-dimensional background numerics and the reproducibility infrastructure developed for this project are sound and unaffected. The next useful calculation is an explicit dimensional reduction of the 5D action, carried far enough to determine whether the induced matter coupling is conformal, as assumed here, or of a derivative type that could evade the sign argument. Until that is done, neither a refit of the transfer function nor a move to six dimensions rests on anything.

Data and code availability

All scripts, outputs and figures accompanying this paper are released with it: `verify_baseline.py` (independent CAMB run and observable extraction), `make_figures.py` (all figures), `baseline_verification.json` (numerical results), `pk_baseline.csv` (power spectra and the σ_8 integrand), and `V2.0_BASELINE_LOCKED.md` (the frozen definitions of Section 2). Every number in this paper is regenerated by running the two scripts in order. Repository: <https://github.com/GeometricCosmo/gb-leakage-cmb>.

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A Growth equation and the role of the higher-order terms

A.1 Numerical growth integration

Writing $N = \ln a$ and using $d \ln H / dN = -\frac{3}{2} \Omega_m(a)$ for a flat Λ background,

$$\frac{d^2 \delta}{dN^2} + \left(2 - \frac{3}{2} \Omega_m(a)\right) \frac{d\delta}{dN} - \frac{3}{2} \Omega_m(a) \frac{G_{\text{eff}}(k, a)}{G} \delta = 0, \quad \Omega_m(a) = \frac{\Omega_m a^{-3}}{\Omega_m a^{-3} + 1 - \Omega_m}. \quad (20)$$

We integrate from $a = 10^{-3}$ with $\delta = d\delta/dN = a$ and form $\Delta P/P = (\delta/\delta_{\Lambda\text{CDM}})^2 - 1$ at $a = 1$. Results are in Table 4.

$k [h\text{Mpc}^{-1}]$	0.01	0.10	0.50	1.50	5.0	20
$\Delta P/P, \beta = 0.10$	+0.047	+0.105	+0.143	+0.158	+0.161	+0.162
$\Delta P/P, \beta = 0.22$	+0.244	+0.612	+0.891	+1.007	+1.039	+1.043

Table 4: Fractional change in $P(k)$ at $z = 0$ from Eq. (18), with $m_{\text{eff}} = 1.5 h\text{Mpc}^{-1}$. Positive at every scale, monotonically increasing, saturating.

A.2 Why the cubic, quartic and derivative terms drop out

Expanding Eq. (11) to first order in δr about a homogeneous background $\bar{r}(t)$:

$$\text{from } (\gamma + 3\beta_3)r^2 : 2(\gamma + 3\beta_3)\bar{r} \delta r, \quad (21)$$

$$\text{from } \delta r^3 : 3\delta \bar{r}^2 \delta r, \quad (22)$$

$$\text{from } \alpha r(\dot{r}^2 - (\nabla r)^2) : \alpha \dot{r}^2 \delta r + 2\alpha \bar{r} \dot{\bar{r}} \delta \dot{r}. \quad (23)$$

The first two are proportional to δr with time-dependent coefficients and are absorbed into $m_{\text{eff}}^2 = V''(\bar{r})$. The third contributes a friction-like term $2\alpha \bar{r} \dot{\bar{r}} \delta \dot{r}$, which is negligible in the quasi-static limit where $\delta \dot{r}$ is dropped relative to $(k^2/a^2)\delta r$, and a mass shift $\alpha \dot{r}^2$. None of them couples δr to $\delta \rho_m$. The matter coupling therefore comes entirely from the F^2 term, as used in Eq. (16).

The one caveat worth recording: if $\bar{r}$ evolves rapidly, so that $\alpha \bar{r} \dot{\bar{r}}$ is not small compared with H , the quasi-static approximation itself requires re-examination. For the parameter values considered here $\bar{r}$ is slowly rolling and the approximation holds, but a construction that relied on fast radion evolution would need this checked.

B Reproduction

```
pip install camb numpy scipy matplotlib
python verify_baseline.py      # -> baseline_verification.json, pk_baseline.csv
python make_figures.py        # -> figures/fig1..fig5 .png
```

Runtime is under two minutes on a single core. CAMB version 2.0.4 was used; any version from 1.3 onward should reproduce σ_8 to better than 10^{-3} . The quadrature in Eq. (2) is Simpson's rule on 4000 logarithmic points over 10^{-4} – $100 h\text{Mpc}^{-1}$; halving the resolution changes σ_8 in the fifth decimal.

C Relationship to earlier versions

The v1.9.0 conclusion, that the mechanism is observationally excluded, was closer to correct than the v1.9.1 revision that replaced it. The intervening change of sign in the project's own conclusions traces to the two convention errors identified in Section 3.

Version	Principal claim	Status after this paper
v1.8.2	Exponential $T(k)$ resolves small-scale abundances	superseded by v1.9.x
v1.8.3	$\sigma_8 = 0.76$ from the stated $T(k)$	incorrect; not reproducible
v1.9.0	S_8 /Lyman- α mutually exclusive; mechanism excluded	directionally correct
v1.9.1	Power-law $T(k)$ passes S_8 and Lyman- α	retracted (Sec. 3)
v2.0	Baseline fails; mechanism falsified	this paper

Table 5: Version history and current status.

D Errata

The following specific statements in earlier releases should not be cited.

1. “ $S_8 = 0.829$ passes the weak-lensing constraint.” It does not; the correct value is 0.8315 and it lies 1.7σ above the target.
2. “The mechanism passes S_8 while failing σ_8 .” Not attainable; $S_8 = 1.025 \sigma_8$ identically.
3. “Lyman- α ratio = 0.891, within bounds.” Computed with $P_{\text{mod}} = TP$ rather than T^2P , and averaged over sampling points at which $T \equiv 1$. The correct four-point value is 0.8125 and the suppression reaches 70% at $k = 5 h \text{Mpc}^{-1}$.
4. “Gradient screening fails by 10^{42} .” Category error; withdrawn. See Section 6.
5. “The holographic derivation fixes $T(k)$ uniquely.” The asymptotic index has holographic motivation; the break scale k_c and the low- k behaviour are fitted.

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